

REDROB AI DISCOVERY & RANKING CHALLENGE

SUBMISSION

1. Team Information

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Problem Statement: Recruiter efficiency is hindered by high volumes (100,000+ profiles) and "trap" profiles, making manual filtering unsustainable.

2. Solution Overview

Q: What is your proposed solution?

A: An automated, high-performance Ranking Engine that filters noise (honeypots) and ranks candidates based on semantic job-description matching and behavioral engagement signals.

Q: What differentiates your approach from traditional candidate matching systems?

A: Unlike keyword-based systems, our solution employs behavioral signal analysis to identify fake/bot profiles and uses semantic ranking to prioritize active, high-quality AI talent.

3. JD Understanding & Candidate Evaluation

Q: What are the key requirements extracted from the JD?

A: Expertise in LLM orchestration, RAG pipelines, production-scale Machine Learning, and complex Retrieval Systems.

Q: Which candidate signals are most important for determining relevance? / How does your solution evaluate candidate fit beyond keyword matching?

A: We evaluate signals like recruiter engagement and profile activity duration to ensure real-world relevance.

4. Ranking Methodology

Q: How does your system retrieve, score, and rank candidates?

A: We use a custom Python pipeline: 1) Honeypot filtering (anomaly detection), 2) Semantic JD scoring (competency mapping), and 3) Behavioral re-ranking (engagement metrics).

Q: What models, algorithms, or heuristics are used?

A: Python-based heuristic engines for fast processing and custom semantic matching logic for high-precision ranking.

Q: How are multiple candidate signals combined into a final ranking?

A: Signals are combined using a weighted scoring algorithm that prioritizes technical competency followed by behavioral activity indicators.

5. Explainability & Data Validation

Q: How are ranking decisions explained?

A: Decisions are based on transparent, deterministic scoring criteria linked directly to JD requirements.

Q: How do you prevent hallucinations or unsupported justifications?

A: The system uses deterministic scoring based on verified JD mappings, avoiding generative "reasoning".

Q: How does your solution handle inconsistent, low-quality, or suspicious profiles?

A: Suspicious profiles are flagged and removed using behavioral signal analysis, identifying anomalous patterns like high response rates with zero connectivity.

6. End-to-End Workflow

Q: What is the complete workflow from JD input to ranked candidate output?

A: Data Ingestion → Honeypot/Trap Filtering → Semantic JD Scoring → Behavioral Engagement Re-ranking → Top-100 Candidate Output.

7. System Architecture

Architecture: A highly optimized, custom Python-based ranking pipeline designed for modularity and high-speed processing.

8. Results & Performance

Q: What results or insights demonstrate ranking quality?

A: The system successfully filters out bot-trap profiles while surfacing candidates with verified RAG/LLM experience.

Q: How does your solution meet the challenge's runtime and compute constraints?

A: The architecture is built for efficiency, ensuring full compliance with the 16GB RAM and 5-minute execution limit.

9. Technologies Used

Q: What technologies, frameworks, and tools were used and why were they selected for this solution?

A: Python (for efficient data manipulation and heuristic scoring) and custom semantic matching logic for high-precision ranking.

10. Submission Assets

GitHub Repository: <https://github.com/sarkarialertportal/redrob-ai-ranker>

Submission File: Completed CSV submission file